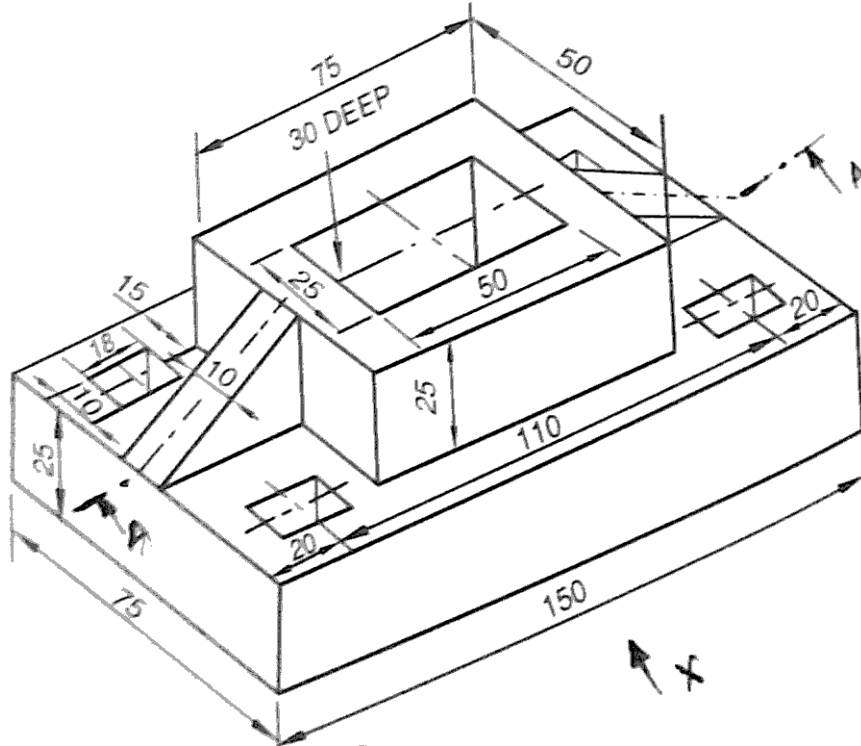


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**End Semester Examination – Summer 2019****Course: B. Tech in Mechanical Engineering****Sem: III****Subject Name: Machine Drawing & CAD****Subject Code: BTMEC304****Max Marks: 60****Date: 31-05-2019****Duration: 3 Hr.****Instructions to the Students:**

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/ CO)	Marks
Q. 1 Solve the following. (Any TWO)		12
A) Draw the conventional representation of	01	06
a) Leaf spring with eye b) Splined shaft c) Tension spring		
B) Draw one example of following sections	03	06
a) Half section b) Removed section		
C) Fig shows a block, Draw i) Sectional Front view along A-A ii) Top view	03	06

**Q.2 Draw the free hand sketch of the following. (Any THREE)****A) Single riveted (Double straps) butt joint.**

02

12**04****B) Castle nut**

02

04

- 02 04

12

- 02

12

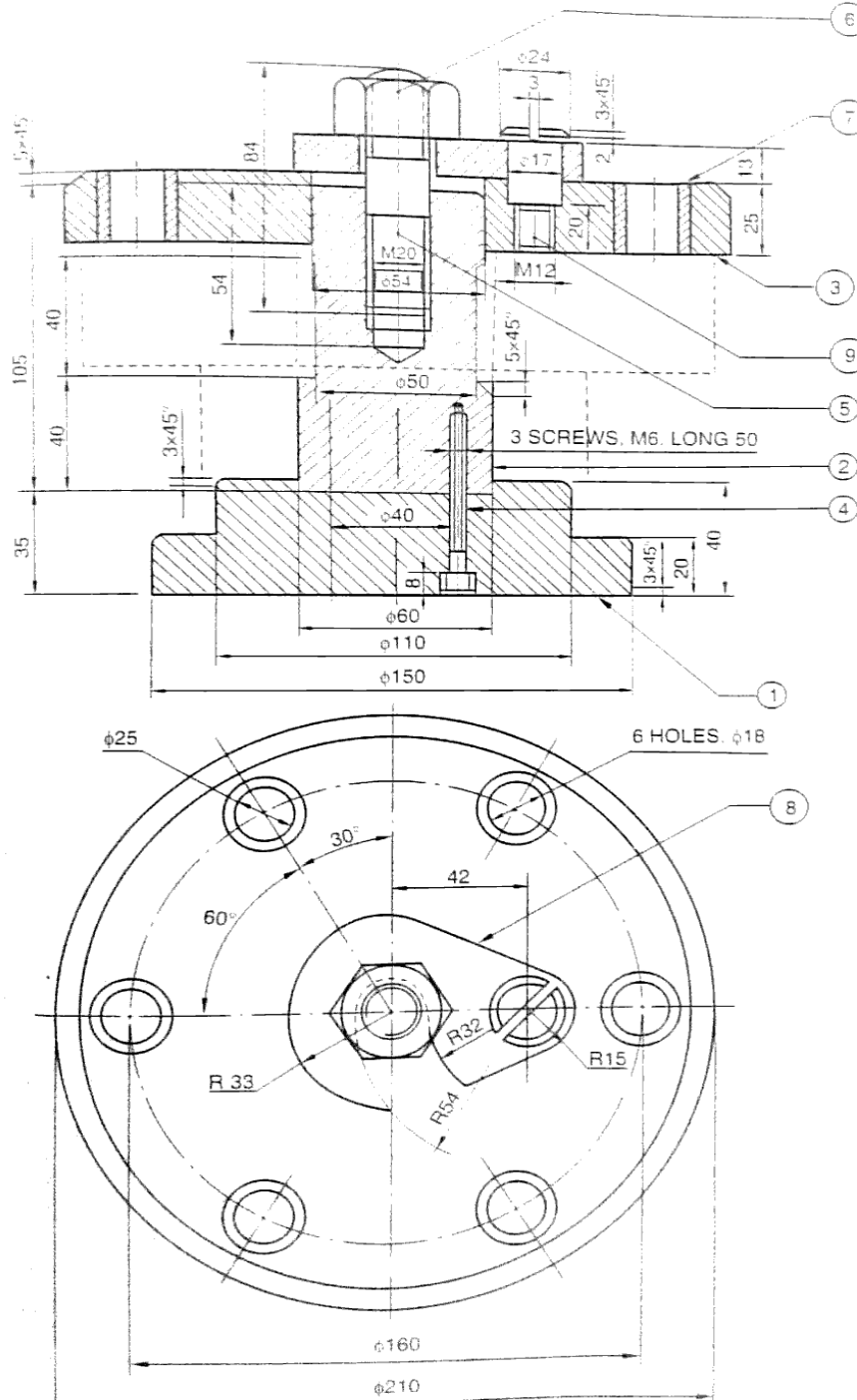
- 03

No.	Name	Matl	Qty
1	Body	Brass	1
2	Cover	Brass	1
3	Valve seat	Bronze	1
4	Valve	Brass	1
5	Stud with nut	MS	6
6	Set screw	MS	1

- B) The fig. shows Assembly of “Drilling Jig”. Draw the Details of part number 01, 02, 03, 04, 05.

Draw the Front view of part number 01, 02, 04, 05.

Draw the Front view and Top View of part number 03.



Part list

PART NO.	NAME	MATERIALS	NO. OFF
1	BASE PLATE	C.I.	1
2	STEM	M.S.	1
3	JIG PLATE	C.I.	1
4	SCREW	M.S.	3
5	STUD	M.S.	1
6	NUT	M.S.	1
7	BUSH	A.S.	6
8	LATCH WASHER	M.S.	1
9	SCREW	M.S.	1

Q. 5 Solve the following.

12

A) The hole size is $\varnothing 20^{+0.018}_{-0.000}$ and shaft size is $\varnothing 20^{+0.023}_{+0.012}$ determine the type of fit.
Dimensions are in mm.

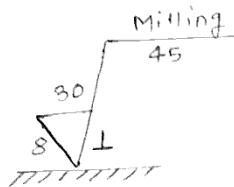
01

04

B) State the meaning of every term shown in following symbol.

02

04

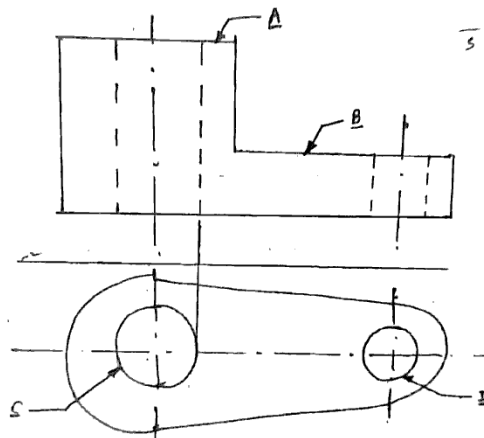


Redraw the following fig. and show the following items on it.

02

04

- C)
- Show the Parallelism between hole C and D with 0.1 mm.
 - Flatness of surface B is within 0.08 mm.



Q. 6 Solve the following

12

- Write advantages of CADD
- Write short note on Auto LISP programming
- List Commands in Draw and modify in Auto CAD.

01

04

01

04

02

04

END